

Project Management Planning

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Section 1: Project Management Software

We have chosen Jira for our project management software. The instructional staff can access our Drone Capstone Team in Jira through the link below. As requested, we have populated our Jira space with the following information for each task: planned start dates, planned completion dates, personnel assigned, and the estimated person-hours needed for completion.

<https://drone-capstone.atlassian.net/?continue=https%3A%2F%2Fdrone-capstone.atlassian.net%2Fwelcome%2Fsoftware%3FprojectId%3D10000&atlOrigin=eyJpIjoiMzk2MTU0YTYxNWMzNDFlZTk0ZDhhNzdhZGQ2YzQ4MTciLCJwIjoiamlyYS1zb2Z0d2FyZSJ9>

Section 2: Person-Hours Available

See the table below for the person-hours available for each team member per week, as well as the total and the weekly average per person for the semester. For our purposes, each week starts on a Monday and ends on a Sunday. Note the general minimum of 12 hours per week for Hannah, Reagan, Yordanos, and Samuel, and the minimum of 15 hours per week for Kevin. The week of Thanksgiving is an exception for a couple of team members, but a minimum of 10 hours of work will still be completed by Hannah and Reagan that week, as opposed to their typical 12 hours. This results in the following total semester hours available for each team member: 130 hours for Hannah and Reagan, 165 hours for Kevin, and 132 hours for Yordanos and Samuel.

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Section 3: Project Backlog/Features List

See the table below for a comprehensive list of all functionalities and tasks needed to implement all of the chosen user stories for both the minimum viable product and the stretch goal features. Note that while some tasks are owned by multiple people in this table format, the individual number of person-hours expected for each team member is provided. Further, in our Gantt chart and our sprint backlog to follow, shared tasks are split into individual entries, with one per person.

ID	User Story	MVP / Stretch	Function	Task	Owner	Priority	Est. Implementation Time	Acceptance Criteria
1. Web and desktop apps, manual flight control, and live footage streaming								
1.1	As a user, I want web and desktop apps with clear, labelled pages for each feature, so that I can find what I need quickly.	MVP	Web app design	Create the first draft of a website with public pages and an operator area (Live, Flights, Zones, Compare, Plan, Settings)	Samuel	High	10 hours	Every web app page is reachable from the header or sidebar and clearly labelled
			Desktop app design	Create the first draft of a desktop app with a sidebar including Home, Control, Sessions, Set Up, and Drone Wi-Fi	Samuel	High	6 hours	Every desktop app page is reachable from the header or sidebar and clearly labelled
1.2	As a user, I want to create an account and sign in, so that only authorized members of the team can see flight data and fly the drone.	MVP	User login	Create sign-up and sign-in features, using one account for both the web and desktop apps	Samuel	High	1 hour	Users can sign up and sign in without errors
			User access	Create an operator role and enforce access in the database	Samuel	High	1 hour	Only a signed-in operator can view flight data or fly the drone
1.3	As an operator, I want to fly the drone within the inspection space manually from my laptop with smooth, stable control, so that I can position it and test it by hand.	MVP	Manual flight	Enable keyboard control of the drone in the desktop app (i.e., the flight agent runs a 50 Hz control loop, holds the spot when keys are released, and lands if the app goes quiet for 0.5 s)	Samuel	High	12 hours	The drone climbs, moves, and holds its position when the corresponding keys are released
1.4	As an operator, I want to see live sensor readings during a flight, so that I can watch the drone and the equipment in real time.	MVP	Live data	Configure live sensor windows in the desktop app (recorded 10 times per second)	Samuel	High	3 hours	Temperature, pressure, battery, and position update on desktop app screen while flying
			Live data	Configure a live page on the website (recorded 10 times per second)	Samuel	High	3 hours	Temperature, pressure, battery, and position update on web app screen while flying
1.5	As an operator, I want to see the camera feed live, so that I can see what the drone sees.	MVP	Live footage	Configure a Camera tab in the desktop app that shows a live feed, where the set-up page installs the camera software on a newly connected drone and joins it to the Wi-Fi	Samuel	High	6 hours	The camera tab shows the drone's current view while it is connected (still frames of ~3.7 per second is acceptable at a minimum)
1.6	As an inspector, I want past flight sessions saved and made comparable (e.g., previous week, previous month), so that I can spot changes over time and plan machine maintenance.	MVP	Session history	Create a Flights page that saves a session history that includes raw readings and charts from completed flights	Samuel	High	2 hours	Collected data from any past flights can be opened and viewed
			Flight comparison	Create a Compare page for comparing up to four flights at a time	Samuel	High	1 hour	Collected data from up to four flights can be compared side-by-side

ID	User Story	MVP / Stretch	Function	Task	Owner	Priority	Est. Implementation Time	Acceptance Criteria
1.7	As an operator, I want the app to check the drone before every flight, so that it never takes off in an unsafe state.	MVP	Flight safety	Check the drone identity, telemetry, motors, battery, decks, base stations, and settled position, writing every action into an audit trail	Samuel	High	2 hours	Check success is reported; check failure blocks take-off and explains the reason in plain words
1.8	As a new team member, I want to install the system with one command on Mac, Windows, or Linux, so that anyone can run it.	MVP	Installation	Bundle the desktop app with the flight agent so that no additional Python set-up is needed, instead relying on a single install command	Samuel	High	8 hours	A fresh machine installs the software and signs in after following the one command
1.9	As an inspector, I want flagged equipment (abnormal or needing further inspection) highlighted in the app with its alert, so that I can review and follow up.	MVP	Flagging	Display the pipeline's alerts for flagged inspection points on the dashboard	Samuel	High	8 hours	Flagged equipment stands out on the dashboard and is accompanied by alert information
			Site design	Allow the user to open and review each alert with ease	Samuel	High	4 hour	Opening an alert shows the sensor reading or image and the reason for the flag
1.10	As an operator, I want to give the system commands in plain English, so that I do not need to learn the interface.	Stretch	LLM assistant	Add an LLM component that turns easily understandable English language commands into app actions.	Samuel, Reagan	Low	12 hours (each)	A plain English request (e.g., "show last week's flagged equipment") returns the correct result
2. Autonomous flight and navigation of the mini-drone								
2.1	As an operator, I want the drone to take off, hover at a set height and land by itself, so that a mission can start and end without anyone piloting it.	MVP	Autonomous flight	Program drone to takeoff from landed position and rise into the air. Once in the air, program the drone to hold position for 10 seconds. After time is reached, program the drone to land safely.	Samuel, Hannah	High	2 hours (each)	The drone rises to 0.3 m above its take-off floor. Once the drone is in the air, it holds for 10 seconds. After 10 seconds, the drone lands with no key pressed.
			Autonomous flight	Measure geometry of the lighthouse and program the drone to fly on it.	Samuel, Hannah	High	2 hours (each)	Flight trace shows it flew on the lighthouse, not the barometer
2.2	As an operator, I want the drone to hold its position and height steadily indoors, so that every reading and image is taken from the same spot.	MVP	Stable flight	Confirm the Lighthouse deck is detected; measure base-station geometry; confirm the position estimate settles.	Samuel, Hannah	High	5 hours (Hannah), 10 hours (Samuel)	Pre-flight check reports the estimate settled under 2 cm. Drift stays under 10 cm during a 10s hover.
2.3	As an operator, I want a programmed virtual fence around the flight area, so that the drone can never leave the safe space.	MVP	Stable flight	Create a Geofence (± 2.0 m, 1.0 m ceiling, checked every 0.1 s in flight; a breach lands the drone.	Samuel	High	3 hours	A route with a point outside the fence is refused before take-off. Crossing the fence in flight lands the drone.

ID	User Story	MVP / Stretch	Function	Task	Owner	Priority	Est. Implementation Time	Acceptance Criteria
2.4	As an operator, I want the drone to fly a predefined inspection route from one marked point of interest to another by itself, so that it can reach a piece of equipment to inspect.	MVP	Autonomous flight	Program drone to take off, go to point A, then go to point B (1m apart)	Samuel, Hannah	High	4 hours (Hannah), 7 hours (Samuel)	Drone arrives within 10 cm of Point B, three runs in a row
			Autonomous flight	Autonomously land drone using the mission waypoint model already in the agent (without manual piloting)	Samuel, Hannah	High	4 hours (Hannah), 7 hours (Samuel)	Once at point B, holds 5s and lands with no manual input, three runs in a row.
2.5	As an operator, I want to plan an inspection route by picking the points of interest in order, so that the drone visits every piece of equipment.	MVP	Route planning	Create a mission planner on the website that builds and validates the route.	Samuel	High	5 hours	Mission planner is visible and interactable on the website that can build and validate a route
			Route planning	Program the flight agent so that it can pick up routes queued on the website.	Samuel, Reagan, Kevin	High	10 hours	A route queued on the website is flown by the desktop app. Points outside the fence are rejected.
2.6	As an operator, I want the drone to return to base and land safely when the mission ends, so that it is ready for the next run.	MVP	Autonomous flight	Add a final return-to-base leg to every mission and land on the starting point.	Hannah, Samuel	High	6 hours (each)	After the last point, the drone lands within 20 cm of where it took off.
2.7	As an operator, I want the drone to fly back to base and land safely when its battery runs low, so that it never drops out of the air mid-mission.	Stretch	Autonomous flight	Define a threshold for low battery and program drone to detect when its low	Hannah, Yordanos	Medium	10 hours (Hannah), 8 hours (Yordanos)	Threshold should be <30 and drone should report when it hits threshold
			Autonomous flight	Program the drone to return to base when threshold is detected and land	Hannah, Yordanos	Medium	14 hours (Hannah), 12 hours (Yordanos)	When the drone reports low battery it stops the mission, returns to base, and lands within 20 cm of where it took off
2.8	As an operator, I want the drone to avoid collisions with walls, people and equipment, so that it does not collide during a mission.	Stretch	Obstacle avoidance	Fit a range sensor for live detection	Kevin	Low	14 hours	A route through a known obstacle is refused. The drone stops before an unexpected obstacle.
			Obstacle avoidance	Program basic obstacle detection. Routes are already checked against a map of known obstacles.	Kevin	Low	16 hours	A route through a known obstacle is refused. The drone stops before an unexpected obstacle.
3. Data collection								
3.1	As an inspector, I want temperature and pressure recorded at inspection points throughout every flight, so that I can see the conditions around each piece of equipment.	MVP	Temperature and pressure sensor data collection	Write raw and corrected temperature, pressure, and position data (recorded 10 times per second, with temperature and pressure captured via specific sensors) to a file	Samuel	High	2 hours	Every flight produces a file with timestamped temperature, pressure, and position rows, including proper units in addition to the data
3.2	As an inspector, I want grayscale images of inspected equipment from	MVP	Camera data collection	Get the AI deck camera to send 324×244 grayscale frames over Wi-Fi (about 3.7 per second), then	Samuel	High	2 hours	A session's frames are saved on the laptop and can be viewed later on the website

ID	User Story	MVP / Stretch	Function	Task	Owner	Priority	Est. Implementation Time	Acceptance Criteria
	the drone's camera, so that equipment can be checked visually.			saving the frames to disk and uploading them				
3.3	As an inspector, I want the collected data wirelessly sent off the drone to an external program during or immediately after flight, and then stored safely so that nothing is lost when a flight ends.	MVP	Data transfer and storage	Transfer collected data from the radio to the laptop (file written first), then upload to the database (retrying if needed until it succeeds).	Samuel	High	2 hours	Flight data appears on the website after the flight, even if the internet was down during it
3.4	As an inspector, I want the drone to capture images and readings while it holds at each inspection point, so that every piece of equipment is recorded the same way.	MVP	Consistency in data collection	Trigger the consistent capture of images and readings during the holding time at each inspection point	Kevin	High	16 hours	At least 10 frames and 5 seconds of readings are saved for every inspection point on the mission route
3.5	As an inspector, I want every collected reading and image tagged with a timestamp and associated inspection point ID, so that I know exactly where and when it was taken.	MVP	Labeling in data collection	Record the timestamp and drone position with every recorded reading	Samuel	High	3 hours	Every reading captured contains a timestamp and drone position data
		MVP	Labeling in data collection	Add an inspection point ID associated with each data collection attempt (i.e., while the drone holds at an inspection point)	Kevin	Low	6 hours	Every reading captured at a point carries that point's unique ID
4. Data processing, anomaly detection, mission replanning, and alerts								
4.1	As a model developer, I want labeled images of the test equipment in normal and faulty states, so that a model can learn the difference.	MVP	Training data	Photograph the test machine in stable and faulty conditions, from several angles and positions	Hannah, Yordanos	High	2 hours (Hannah), 5 hours (Yordanos)	A labeled image set with both classes (normal and faulty), from several angles and positions
4.2	As an inspector, I want erroneous sensor readings removed, so that a faulty sensor is not mistaken for faulty equipment.	MVP	Sensor data preprocessing	Design a method for identifying outliers and gaps when a sensor misbehaves, including marking and removing them	Kevin	High	8 hours	Erroneous values planted in a flight data file are flagged, while correct rows are left untouched
4.3	As an inspector, I want clearer images, so that even small defects are visible.	MVP	Image data preprocessing	Find and apply a model or program that increases the quality and resolution of the 324×244 grayscale frames	Kevin, Reagan	Medium	12 hours (each)	Before vs. after images on real frames captured by the drone show visibly more detail
			Image data preprocessing	Find and apply a model or program that can colorize pictures after the fact for better visual clarity	Kevin	Low	10 hours	Grayscale images are converted to show the color of objects accurately
4.4	As an inspector, I want a model that looks at an image and can define it as faulty or not, so that defects are found automatically.	MVP	ML classification	Train a simple machine learning classification model on labeled images (faulty vs. not faulty)	Reagan	High	12 hours	Accuracy reported on a held-out test set, with the review of misclassified examples
4.5	As an inspector, I want each flight's images and sensor readings to go	MVP	Data pipeline	Establish one command to load a flight's sensor file and frames from	Kevin, Reagan	High	10 hours (each)	A single command turns a recorded flight into a result with

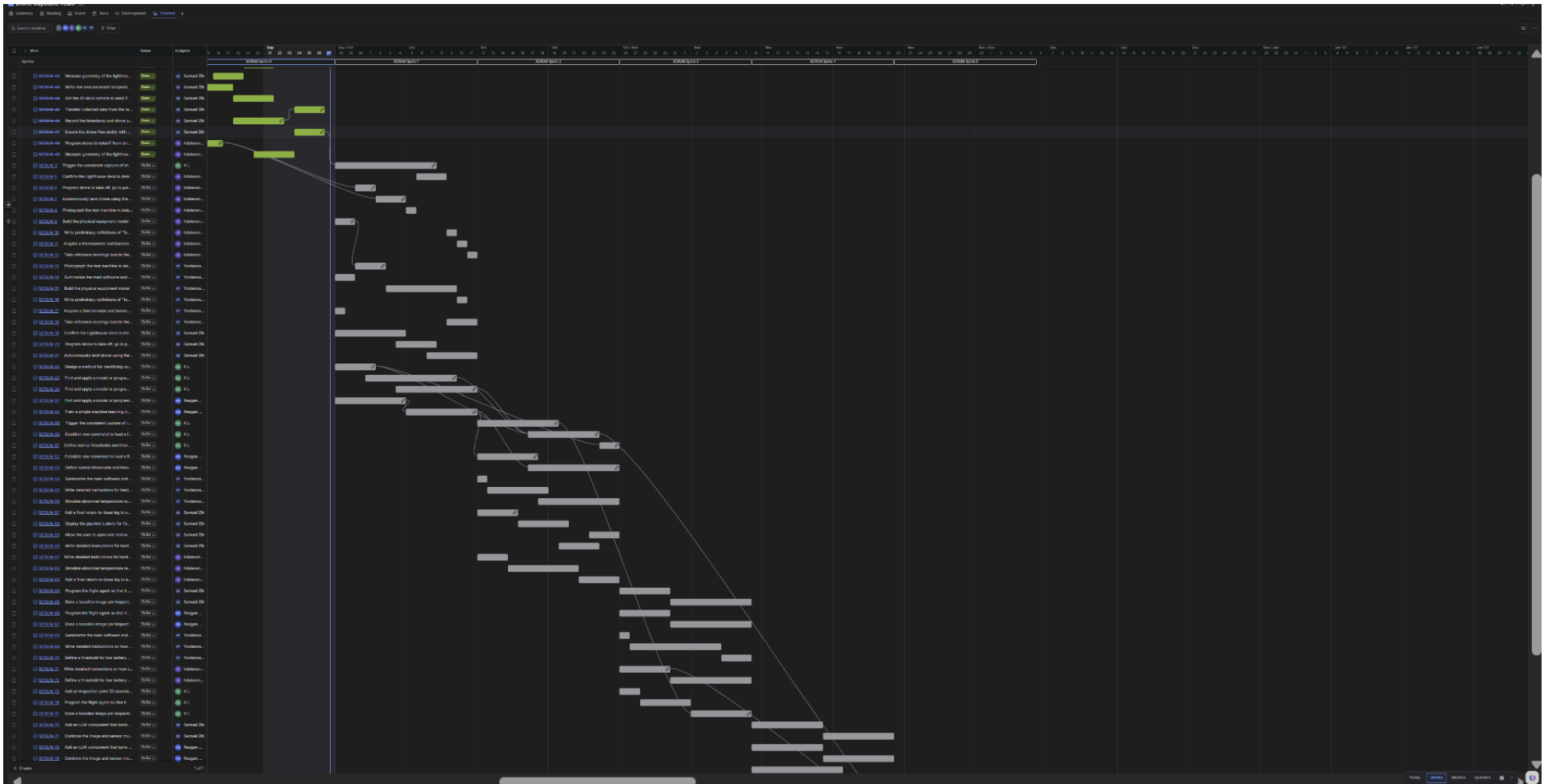
ID	User Story	MVP / Stretch	Function	Task	Owner	Priority	Est. Implementation Time	Acceptance Criteria
	through one pipeline (clean, enhance, classify, interpret), so that every flight ends with a verdict for each piece of equipment without manual steps.			storage, run the data pipeline steps in order (clean, enhance, classify, interpret), then save the result.				a verdict for each inspection point.
4.6	As an inspector, I want an external program to compare sensor readings against defined normal operating thresholds, so that overheating, pressure changes, or other found anomalies are flagged, with the reason so that I can act on it and review it later.	MVP	ML classification / alerts	Define normal thresholds and then train a machine learning classifier on sensor data, implementing a model that flags any deviations and highlights suspicious data offline, with the reason provided	Kevin, Reagan	High	4 hours (Kevin), 14 hours (Reagan)	An anomaly is flagged and an alert is displayed to the user, alongside the original reading or image (e.g., box heated with a hand warmer is flagged; while unheated box is not)
4.7	As an inspector, I want visual changes/abnormalities in captured images flagged against a baseline reference image, so that a change in the equipment is noticed even without a trained fault class.	Stretch	Visual anomaly	Store a baseline image per inspection point and compare each new image against it	Reagan, Kevin, Samuel	Medium	14 hours (each)	A visible change to a box is flagged, while the same box unchanged is not
4.8	As an inspector, I want the model to use images and sensor data together, plus the equipment's history, so that its verdicts are more reliable.	Stretch	ML classification	Combine the image and sensor models with past readings for the same equipment when classifying equipment condition	Reagan, Samuel	Low	12 hours (each)	This combined model performs with at least as much accuracy as either sub-model alone when applied to the test set
4.9	As an operator, I want anomalies flagged during the flight, not after it, so that the mission can react right away.	Stretch	Real-time detection	Design and run the machine learning classifier on live data as it arrives (online)	Samuel, Reagan, Kevin	Medium	12 hours (Samuel), 12 hours (Reagan), 15 hours (Kevin)	An anomaly is flagged and an alert is displayed to the user while the drone is still at that point (e.g., a heated box is flagged in real-time)
4.10	As a user, I want the drone to alter its mission in real-time and inspect a flagged point again (second, closer pass) when a sensor finds an issue, then return to its base.	Stretch	Revisit plan / second inspection	Prepare the steps required for the drone to revisit a flagged location, collect new data, and return to the route	Yordanos, Hannah	Medium	10 hours (Yordanos), 12 hours (Hannah)	A valid warning triggers re-navigation, while old, repeated, or erroneous warnings do not; new data is collected upon revisit and the drone proceeds to the next waypoint; any failed checks are recorded
5. Documentation								
5.1	As a user, I want documentation of the project, so that I can understand how it was implemented.	MVP	Project documentation	Summarize the main software and hardware decisions, modifications, current status, and challenges encountered throughout the project by the group	Yordanos	High	2 hours (each sprint)	Clear and complete notes on the main choices and problems, with unresolved issues marked

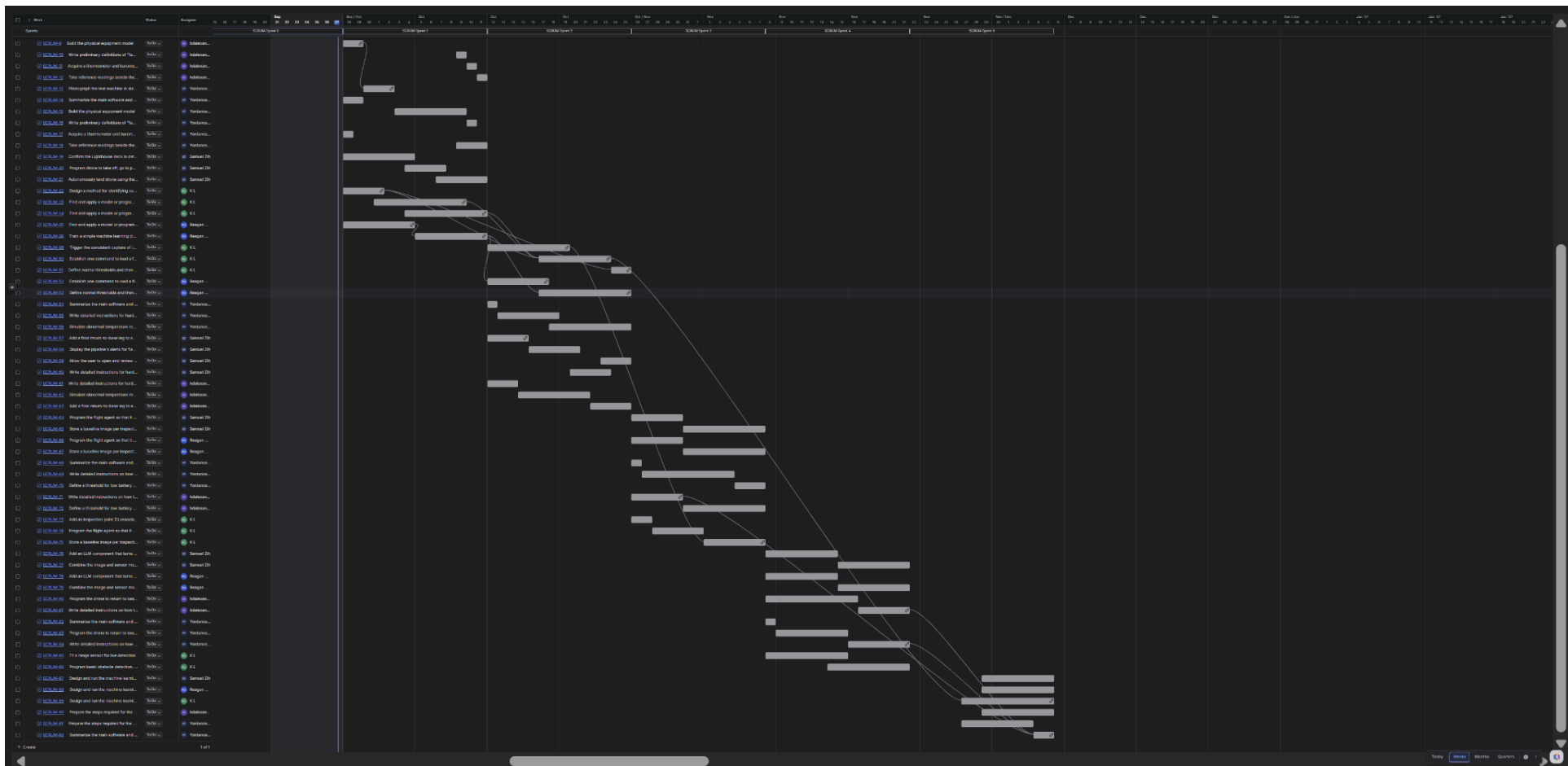
ID	User Story	MVP / Stretch	Function	Task	Owner	Priority	Est. Implementation Time	Acceptance Criteria
5.2	As a user, I want set-up and operating instructions for the mini-drone system, so that I can prepare the drone for a mission.	MVP	Setup guide	Write detailed instructions for hardware and software setup, and the safety checks before a flight	Samuel, Yordanos, Hannah	High	6 hours (each)	A person outside of the team can follow the instructions and connect to the drone properly and safely
5.3	As a user, I want simple and clear instructions on how to operate the drone safely.	MVP	Operating guide	Write detailed instructions on how to start, monitor, land, and end a mission with the drone, including troubleshooting fixes for common problems	Yordanos, Hannah	High	14 hours (each)	A user can run a mission successfully using the guide, and can reference safe actions for any failed missions
5.4	As a user, I want a defined scheme of markers and waypoints, so that I can operate the drone in a new indoor space.	MVP	Marker and waypoint guide	Write detailed instructions on how to name, place, and test markers and waypoints in a new indoor space	Yordanos, Hannah	Medium	10 hours (each)	Ensure that each marker has one waypoint inside the flight area, and that missing, repeated, or blocked points are not accepted
6. Testing and experiment design								
6.1	As the team, we want simulated equipment built from boxes, with a definition of "faulty", so that we can test inspection without real machinery.	MVP	Experiment design	Build the physical equipment model	Hannah, Yordanos	High	5 hours (Hannah), 10 hours (Yordanos)	Test boxes are placed inside the flight area
			Experiment design	Write preliminary definitions of "faulty".	Hannah, Yordanos	High	1 hour (each)	Each test box has a written normal and faulty definition
6.2	As the team, we want independent thermometer and barometer readings, so that we can confirm the drone measured a real change.	MVP	Experiment design	Acquire a thermometer and barometer	Hannah, Yordanos	High	1.5 hours (each)	Both thermometer and barometer have been acquired and tested
			Experiment design	Take reference readings beside the test boxes.	Hannah, Yordanos	High	1.5 hours (Hannah), 4.5 hours (Yordanos)	Reference readings are recorded next to the drone's readings for the same spot and time
6.3	As the team, we want to create abnormal temperatures on purpose, so that we can check that the system catches them.	MVP	Experiment design	Simulate abnormal temperature readings using a hand warmer.	Hannah, Yordanos	Medium	16 hours (Yordanos), 12 hours (Hannah)	The drone's reading near the heated box differs from the unheated one by a measurable amount.
6.4	As a contributor, I want the drone confirmed ready for flight and data collection, so that test time is not wasted.	MVP	Manual flight	Ensure the drone flies stably with manual controls; run automated tests on every change.	Samuel	High	4 hours	Manual flight works; the flight agent's 695 automated tests pass

Section 4: Gantt Chart / Product Roadmap

See the screenshots below for a Gantt chart that compiles our tasks across all sprints. Note that several tasks have already been completed prior to the start of Sprint 1, so they are grouped under Sprint 0 for documentation purposes. We have included two screenshots to represent the static image, but we recognize that the content is not very valuable in this form. As Jira would not allow us to export the Gantt chart as a single image without significant formatting issues, please follow the link that follows for the full chart. To view the full Gantt chart with delineated tasks, under “Sprints”, click the “>” for “Expand hierarchy” just to the left of “SCRUM-94.” Scroll and zoom out as needed to view a larger portion of the chart at once. The estimated number of person-hours can be found by clicking on an individual task, and viewing the “Story point estimate” field, which we have used as a dedicated storage location for the estimated person-hours (not to be confused with actual story points). Inter-task dependencies are also indicated with connecting lines.

<https://drone-capstone.atlassian.net/jira/software/projects/SCRUM/boards/1/timeline?rangeMode=WEEKS>.





Section 5: Sprint 1 Backlog

See the screenshot below for the assignable tasks that will be completed in Sprint 1.

Sprint 1

Item	Task Backlog	Owner	Est. Completion Time (hours)	Status	Actual Completion Time (hours)	Demo
1	Confirm the Lighthouse deck is detected; measure base-station geometry; confirm the position estimate settles.	Hannah	5	To Do		Pre-flight check reports the estimate settled under 2 cm. Drift stays under 10 cm during a 10s hover.
2	Program drone to take off, go to point A, then go to point B (1m apart)	Hannah	4	To Do		Drone arrives within 10 cm of Point B, three runs in a row
3	Autonomously land drone using the mission waypoint model already in the agent (without manual piloting)	Hannah	4	To Do		Once at point B, holds 5s and lands with no manual input, three runs in a row.
4	Photograph the test machine in stable and faulty conditions, from several angles and positions	Hannah	2	To Do		A labeled image set with both classes (normal and faulty), from several angles and positions
5	Build the physical equipment model	Hannah	5	To Do		Test boxes are placed inside the flight area
6	Write preliminary definitions of "faulty".	Hannah	1	To Do		Each test box has a written normal and faulty definition
7	Acquire a thermometer and barometer	Hannah	1.5	To Do		Both thermometer and barometer have been acquired and tested
8	Take reference readings beside the test boxes.	Hannah	1.5	To Do		Reference readings are recorded next to the drone's readings for the same spot and time
9	Summarize the main software and hardware decisions, modifications, current status, and challenges encountered throughout the project by the group	Yordanos	2	To Do		Clear and complete notes on the main choices and problems, with unresolved issues marked
10	Photograph the test machine in stable and faulty conditions, from several angles and positions	Yordanos	5	To Do		A labeled image set with both classes (normal and faulty), from several angles and positions
11	Build the physical equipment model	Yordanos	10	To Do		Test boxes are placed inside the flight area

12	Write preliminary definitions of "faulty".	Yordanos ▾	1	To Do ▾		Each test box has a written normal and faulty definition
13	Acquire a thermometer and barometer	Yordanos ▾	1.5	To Do ▾		Both thermometer and barometer have been acquired and tested
14	Take reference readings beside the test boxes.	Yordanos ▾	4.5	To Do ▾		Reference readings are recorded next to the drone's readings for the same spot and time
15	Confirm the Lighthouse deck is detected; measure base-station geometry; confirm the position estimate settles.	Samuel ▾	10	To Do ▾		Pre-flight check reports the estimate settled under 2 cm. Drift stays under 10 cm during a 10s hover.
16	Program drone to take off, go to point A, then go to point B (1m apart)	Samuel ▾	7	To Do ▾		Drone arrives within 10 cm of Point B, three runs in a row
17	Autonomously land drone using the mission waypoint model already in the agent (without manual piloting)	Samuel ▾	7	To Do ▾		Once at point B, holds 5s and lands with no manual input, three runs in a row.
18	Design a method for identifying outliers and gaps when a sensor misbehaves, including marking and removing them	Kevin ▾	8	To Do ▾		Erroneous values planted in a flight data file are flagged, while correct rows are left untouched
19	Find and apply a model or program that increases the quality and resolution of the 324×244 grayscale frames	Kevin ▾	12	To Do ▾		Before vs. after images on real frames captured by the drone show visibly more detail
20	Find and apply a model or program that can colorize pictures after the fact for better visual clarity	Kevin ▾	10	To Do ▾		Grayscale images are converted to show the color of objects accurately
21	Find and apply a model or program that increases the quality and resolution of the 324×244 grayscale frames	Reagan ▾	12	To Do ▾		Before vs. after images on real frames captured by the drone show visibly more detail
22	Train a simple machine learning classification model on labeled images (faulty vs. not faulty)	Reagan ▾	12	To Do ▾		Accuracy reported on a held-out test set, with the review of misclassified examples